What a Lineage Learns: Thirty-Nine Pre-registered Predictions in a Sealed Artificial World with a River

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Abstract

We report six experiments in river world, a sealed artificial ecosystem in which tabular Q-learning agents live inside an evolving population under a strictly conserved energy budget, on a cylinder of land split by a river that can be crossed only at fords. Nine conditions were run for sixteen seeds each, to 2,000 simulated years and, for four of them, to 10,000: 800,000 seed-years in all. Thirty-nine predictions were written down before the runs they concern and are scored here unedited: 17 held, 5 held in part and 17 failed. Three results stand out. First, the apparent Baldwin-style effect, evolution turning down the learning rate when learned values are not inherited, is a transient: the ratio between arms peaks at 3.7 near year 300 and falls to 1.2 by year 10,000, so inheritance changes how fast the learning rate converges, not where. Second, inheriting learned values is the only route to a signalling habit: a 0.05 energy cost to speak is invisible to one lifetime of learning but is learned by a lineage, and 83% of inheriting agents end up favouring silence against 33% without inheritance. Third, walling the river off entirely produces no divergence in signalling between the banks, because the channel offers a cost gradient rather than an arbitrary convention to diverge on. The failed predictions were the informative ones: each cluster of failures pointed at a place where the model's mechanism differed from its description.

Introduction

Artificial life has long asked how learning within a lifetime and evolution across lifetimes shape each other (Baldwin, 1896; Hinton and Nowlan, 1987; Ackley and Littman, 1991), whether learned knowledge passed directly to offspring helps or harms a population (Ackley and Littman, 1994), and whether isolated populations of communicating agents drift into distinct conventions (Steels, 1995; Kirby, 2002). Persistent multi-agent worlds (Ray, 1991; Yaeger, 1994; Suarez et al., 2019) make these questions testable over long horizons, and open-endedness gives them urgency (Bedau et al., 2000; Packard et al., 2019; Hughes et al., 2024).

Recent work keeps these questions open. Direct inheritance of learned or epigenetic state is again being compared with the Baldwin effect, in evolutionary computation (Benito et al., 2026) and on compositional tasks, where it accelerates learning (Le, 2026). Populations of learning agents have been evolved in large worlds that are never reset (Hamon et al., 2023), cultural knowledge has been shown to accumulate across generations of reinforcement learners (Cook et al., 2024), and novelty, complexity and adaptation have been measured side by side in an open-ended system (Moreno et al., 2024). River world sits where these meet: learners inside an evolving, never-reset population, with the inheritance rule as an experimental switch and a barrier that can isolate two halves of the culture.

These worlds also make it easy to tell a story afterwards. A long run produces dozens of time series, and with enough of them some pattern can always be found and explained. Experimental psychology's answer to the same problem is pre-registration: writing the prediction, and what would falsify it, before the data exist (Nosek et al., 2018). It is rare in artificial life.

This paper applies it to one world. We built river world to join a lifetime learner to an evolving population under a fixed energy budget, wrote a set of predictions for six experiments, ran them, and score every prediction here, including the seventeen that failed. Our contributions are:

- A sealed, energy-conserving world combining tabular Q-learning, a heritable genome that tunes learning, and a geographic barrier, reproducible byte for byte across machines.
- Evidence that the Baldwin-style divergence in learning rate between inheriting and non-inheriting populations is transient, and that its size depends on when it is measured.
- A case of lineage-level learning: a behaviour too costly to show up within one life, acquired by a lineage through inherited value estimates.
- A null result for isolation and signalling, explained

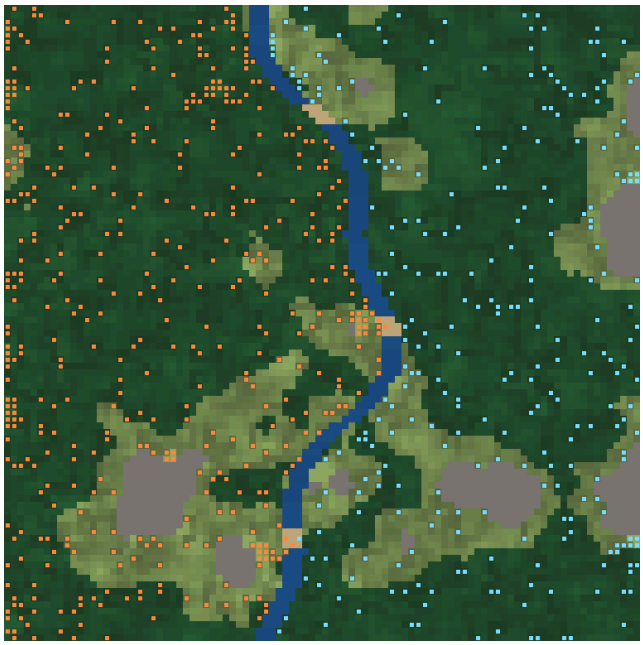


Figure 1: River world after 300 simulated years (one seed, learning inherited). The river runs north to south and wraps round; tan bars are the three fords. Agents are coloured by bank. Paler land holds more nutrients.

by the shape of the signal channel's costs, with the experiment that would test the explanation.

- A full scorecard of 39 pre-registered predictions, and what the failures revealed.

The model

World. The world is a grid of 96×96 cells that wraps north to south and is walled east to west, so it is a cylinder (Figure 1). Terrain is forest, clearings (22% of cells) and impassable rock (5%). A river three cells wide meanders down the length of the cylinder and can be crossed only at three fords. Time runs in ticks: a day is 20 ticks, the last 8 of them night, a season 100 and a year 400.

Energy. The world holds 30,000 energy units, and none is ever created or destroyed; the total is asserted after every tick. Energy sits in four pools: nutrients on the land, the river's suspended load, agents' bodies and detritus. Agents eat nutrients (up to 4 a tick, at most 150 held) and pay 0.5 a tick to rest and 0.9 to move, a quarter more at night. A dead agent becomes detritus, which decomposes back into nutrients at 2 to 5% a tick by terrain, faster in spring and slower in winter. Nutrients diffuse 2% a tick to each land neighbour. The river erodes 0.2% of bank nutrients a tick, carries its load downstream and deposits 1% as silt. Seasonal

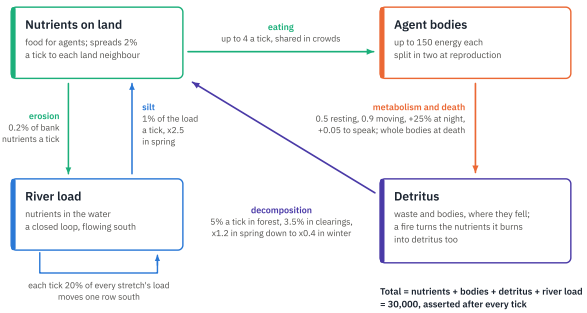


Figure 2: Where the energy goes. Four pools hold all 30,000 units; arrows are the only ways energy moves between them. Events change the rates on the arrows, never the total.

events (spring floods, summer droughts, fires of radius 6) change these rates but never the total (Figure 2).

Agents. Each action pairs a move (stay or one of four directions) with a signal (silent or one of three symbols), 20 actions in all. An agent hears the loudest signal in its eight neighbouring cells one tick later; speaking costs 0.05 energy a tick, whichever symbol is used. Each agent is a tabular Q-learner (Watkins and Dayan, 1992) over 240 states: the direction of the richest neighbouring cell, its energy band, day or night, the signal it hears and whether it is near water. The reward is the change in its own energy; the discount is 0.95.

Evolution. An agent with energy above a heritable threshold splits it with a child. The genome has five genes: lifespan, the reproduction threshold, the mutation scale, the learning rate α (starting at 0.2) and the exploration rate ε (starting at 0.1). Evolution can therefore tune the learning machinery itself: in Hinton and Nowlan (1987) the genome decides which traits are left to learning, whereas here it sets the learning and exploration rates. What the child receives of its parent's learning is set by a switch: copy hands over the parent's Q-table (Lamarckian inheritance of learned values), half hands it over at half strength, and none starts every child blank, so only the genes for learning are inherited. The default is copy.

Method

Experiments. Six experiments each change one thing against the defaults (Table 1). E1 and E2 ask whether learning should be inherited; E3 whether the river is a barrier to culture; E4 whether disturbance keeps the world varied, after the intermediate disturbance hypothesis (Connell, 1978); E5 whether the population is

Table 1: The nine conditions. Anything not named is at the default: 96×96 cells, 30,000 energy units, a 2,000-agent cap, copy inheritance, events on.

Condition	Change	Exp.	Years
base	none	all	10,000
none	inherit none	E1, E2	10,000
half	inherit half	E1, E2	2,000
calm	no events	E4	2,000
e20k	20k energy	E5	2,000
e45k	45k energy, cap 3k	E5	2,000
big	128^2, 53k energy, cap 3.5k	E6	2,000
ford1	one ford	E3	10,000
wall	no fords	E3	10,000

limited by starvation or by crowding; and E6 whether a bigger world sustains more variety. The agent cap sits at about twice the expected population and is scaled with energy so that it never binds.

Runs. Every condition ran sixteen seeds to year 2,000. Four conditions were carried on to year 10,000, chosen by a trend test over years 1,500 to 2,000 rather than by taste: base was still moving, ford1 and wall were E3's arms, and none is the control for E2. That is 800,000 seed-years, run on a single 16-core desktop at about 1.1 simulated years a second per process. Every logged quantity is a per-year population statistic: population, the five gene means, the number of states visited, policy complexity, lineage depth, signal entropy, the Jensen–Shannon divergence between the two banks' signal distributions (dialect_jsd), crossings, the share of the population on each bank and a behaviour-descriptor archive, from which a novelty count is derived in the spirit of the MODES novelty metric (Dolson et al., 2019).

Reproducibility. A seed determines a world. Seeds 1 to 4 of base, regenerated from scratch on the desktop, reproduced a 300-year run of the same seeds made earlier on a laptop with a different processor, on every one of 24 logged columns, byte for byte, except the floating-point residue of the conservation check. A stopped run resumes exactly. Energy drift stays near one part in 10^{12} of the total.

Pre-registration. Twenty-six predictions, each with a number that would make it wrong, were written to a dated file before the first world was created. Four more were added for e45k while it was still running, after e20k had finished, and nine more before the four long runs were carried past year 2,000. None has been edited; the file is archived with the data (Bremner, 2026b).

Table 2: Mean learning-rate gene over the 100 years ending at each year shown, sixteen seeds an arm. The ratio peaks near year 300 and decays.

Year	100	300	1k	2k	5k	10k
copy	0.216	0.168	0.089	0.061	0.044	0.039
none	0.108	0.045	0.034	0.033	0.032	0.033
Ratio	2.0	3.7	2.6	1.86	1.37	1.20

Comparisons use Welch's t over the last 100 years, sixteen seeds a side. Because each comparison tests about twenty-two columns, we treat only $|t| > 4$ as a result.

Results

Inheritance: a transient gap and a permanent habit

The Baldwin gap is a transient. When children start blank, evolution turns the learning rate down quickly; when they inherit their parent's Q-table, it falls too, but later (Figures 4 and 3, Table 2). The ratio of the two arms' mean α, as 100-year means, was 3.7 at year 300, 2.6 at 1,000, 1.86 at 2,000, 1.37 at 5,000 and 1.20 at 10,000, where $t = 2.3$. The none arm reached $\alpha \approx 0.034$ by year 1,000 and has stayed between 0.032 and 0.034 since. Read at year 300, this would be a large, permanent effect of inheritance on learning; read at year 10,000, it is a difference in how long the population takes to reach the same place. Exploration has narrowed less (ratio 2.1 to 1.5) and still separates at $t = 5.7$.

A halfway rule is not a halfway world. The half arm was meant to land between the other two. It does not: it matches copy on states visited and none on the learning-rate gene, and on exploration it lies outside both. The cause is one line of code. A child receives the parent's whole visited mask and every value halved, so the halving compounds down the generations. Reading the checkpoints at year 2,000, a half agent has visited as many states as a copy agent (136 of 240 against 135) but holds values 115 times weaker, weaker than what a blank agent learns for itself in one life. Halving preserves which action is best but not how strongly it is preferred, and evolution tunes the learning rate as though nothing were inherited.

A habit only a lineage can learn. Without inheritance, signal entropy stays at 1.997 bits of a possible 2.0 for all 10,000 years: the four signals are used almost exactly equally, and silence is the favourite of 33% of agents against the 25% chance would give (Table 3). With inheritance, silence is the favourite of 83%. Speaking

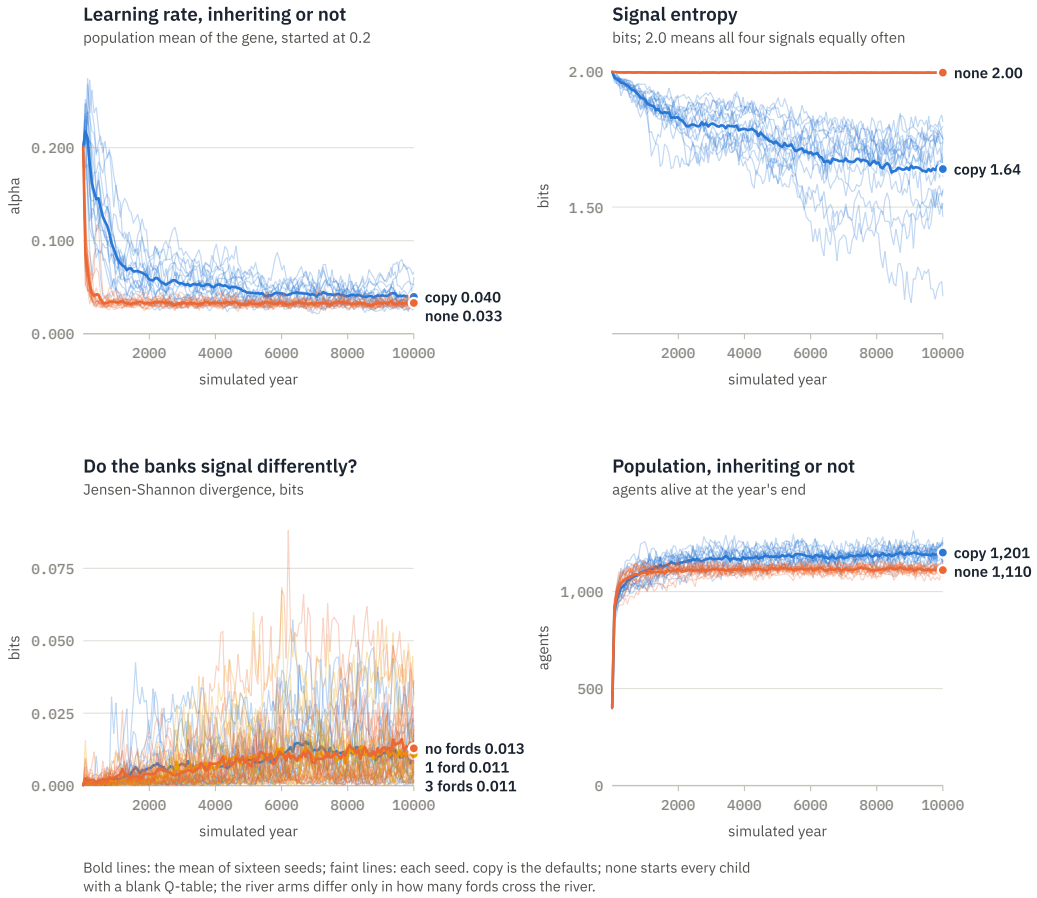


Figure 3: The four conditions carried on to year 10,000, sixteen seeds each; bold lines are the mean, faint lines individual seeds. Top left: the learning-rate gap between copy (blue) and none (orange) closes to within seed noise. Top right: only the inheriting world structures its signalling. Bottom left: the river arms stay together throughout. Bottom right: inheriting learned values supports 8% more agents, permanently.

Table 3: copy against none at year 10,000: means over years 9,901 to 10,000, sixteen seeds a side.

Measure	copy	none	t
Learning rate α	0.0391	0.0326	2.3
Exploration ε	0.0212	0.0145	5.7
Signal entropy (bits)	1.654	1.997	−8.9
Favourite signal is silence	83%	33%	
Population	1,198	1,111	9.8
Generations deep	2,462	1,793	14.4
Nutrients standing on land	4,839	4,155	15.2
Crossings a year	183	318	−4.7

costs 0.05 energy a tick, a cost too small for one lifetime of Q-learning to detect against the noise of feeding, and large enough to show once the estimate is carried down 2,400 generations. This is a behaviour that no individual learns and a lineage does.

Inherited learning feeds more agents. The inheriting world holds 7.8% more agents ($t = 9.8$), a gap present at every 1,000-year checkpoint from year 2,000. It also leaves more nutrients standing and holds more energy in each body, so the extra agents come from grazing better rather than harder. The blank world's standing crop has been flat at 4,153 to 4,155 units since year 2,000; the inheriting world's rose from 4,544 to 4,839 and was still rising when the run stopped.

Isolation does not separate the banks

Walling the river completely cut crossings to zero in every one of wall's 80,000 seed-years to year 5,000, and one ford cut them to a third. Nothing downstream followed (Figure 5). At year 5,000, the pre-registered horizon, dialect_jsd was 0.0086 bits with three fords, 0.0090 with one and 0.0083 with none ($t = 0.12$, wall against three fords). All three arms drift upward together, from

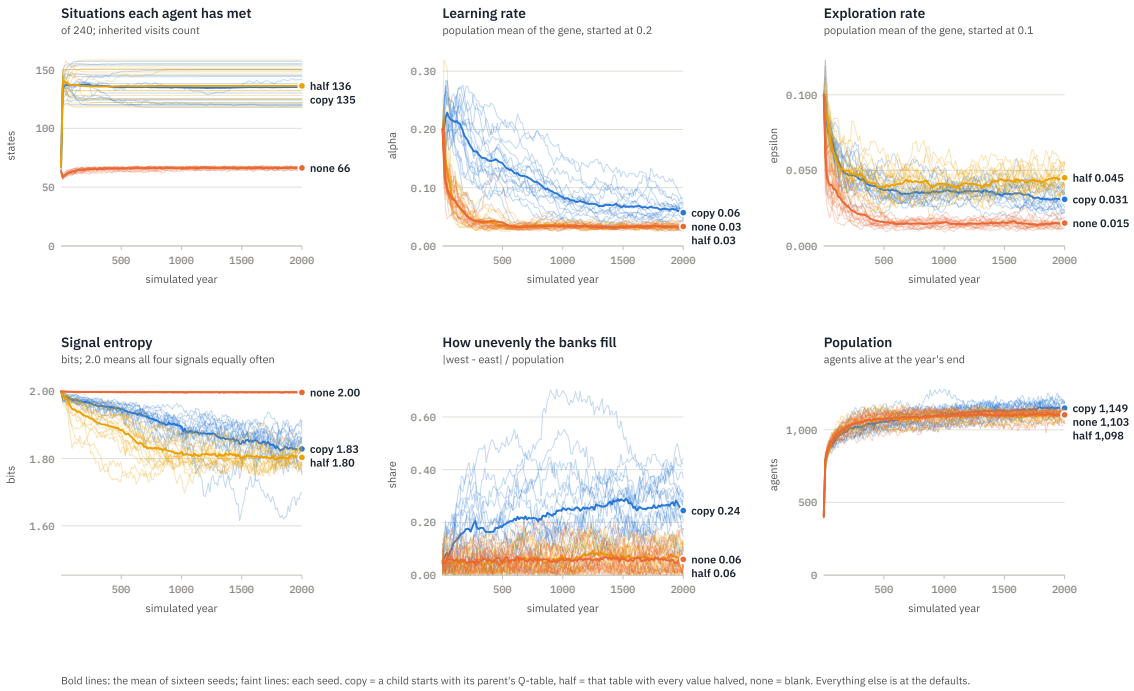


Figure 4: Three inheritance rules over 2,000 years, sixteen seeds each; bold lines are the mean, faint lines individual seeds. The learning-rate gap is real and narrowing, and the middle rule (half) does not produce a middle world.

about 0.001 at year 500, which is a property of the world ageing rather than of its geography. At year 10,000 the values are 0.0098, 0.0100 and 0.0130 ($t = 0.98$). The largest single seed-year in any arm in 10,000 years was 0.096; the pre-registered threshold for a dialect, 0.2, was never approached.

The reason is in what agents say, which is mostly nothing. At year 5,000, 79 to 80% of agents on both banks favour silence with three fords open, and 75 to 76% with none; the three spoken symbols share the remainder almost evenly. Silence is free and the symbols all cost the same, so selection pushes both banks towards silence identically, and among the symbols there is nothing to choose. Where the choice between conventions is arbitrary, a population settles on one by self-organisation (Steels, 1995), and which one depends on its history, so isolated populations are free to settle differently. This channel offers a cost gradient instead, and a cost gradient points the same way on both banks. One measure leans: the walled world's signal entropy is higher (1.752 against 1.654, $t = -2.4$) in the same direction at years 2,000, 5,000 and 10,000, consistent with silent lineages recolonising across fords, but below our bar for a result.

Energy and size: a fixed standing crop

Across a 2.25-fold range of total energy (e20k, base, e45k) the nutrients standing on the land were 0.535, 0.533 and 0.541 units a land cell. Taking 10,000 units out of the world took 9,780 from detritus and none from the land; adding 15,000 put 14,577 into detritus. The land is grazed to the same density every time and the remaining energy waits in the soil. The same constant holds across size: big, with 1.79 times the land and the same energy a cell, stands at 0.536.

So population is more than proportionally sensitive to energy: 584, 1,151 and 2,066 agents, close to $0.059 \times E - 614$, which implies about 10,400 units of overhead before the world supports one agent. A richer world buys more distinct behaviours (archives of 557, 751 and 994) and costs learning: both α and ε fall as the population grows denser and each agent holds less. A bigger world buys only more agents. big holds 1.76 times the population, with behaviour diversity unchanged to three significant figures and its archive 7% larger, inside seed noise ($t = -1.4$), while 1,243 of the archive's 2,048 cells remain unvisited. The space does not fill; the world stops exploring it.

Disturbance, as sized, does nothing

Switching fires, floods and droughts off (calm) changed no logged measure by more than $|t| = 1.9$, and the two

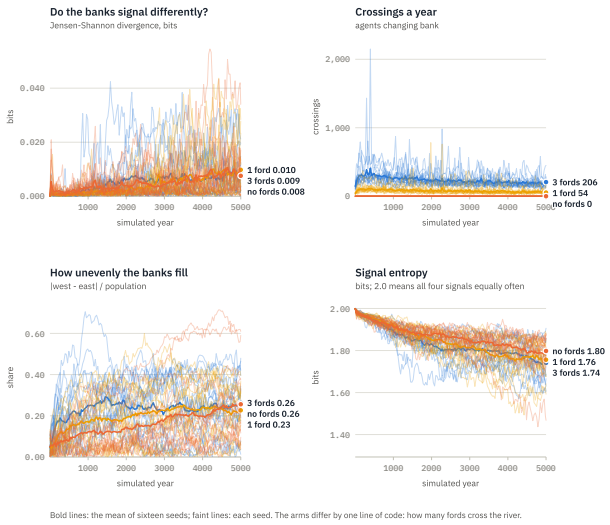


Figure 5: E3, sixteen seeds an arm over 5,000 years. Crossings fall from about 200 a year (three fords) to about 60 (one ford) to none (wall); the divergence between the banks' signalling does not move with them.

Table 4: Thirty-nine pre-registered predictions by when they were written. Two concern hardware throughput rather than the model.

Written	Scored	Right	Partly	Wrong
Before any run	26	9	5	12
Mid-programme (e45k)	4	3	0	1
Before the long runs	9	5	0	4
All	39	17	5	17

closest pointed the wrong way for the hypothesis. The instrument was not asleep: within base, years with an event and years without show the same annual population change, and events add 1 to 3% to its volatility. The reason is arithmetic. A fire burns about 113 of 8,520 land cells and moves roughly 60 units of nutrient into detritus; floods and droughts act on the river's load, which holds 24 of the world's 30,000 units. Every disturbance works on about 0.1% of the energy budget in a world where 25,000 units sit in detritus. E4 is therefore deferred, not answered.

The predictions, scored

Of 39 scored predictions, 17 held, 5 held in part and 17 failed (Table 4). They improved as the evidence behind them grew: 35% of the first set held in full, 75% of the mid-programme set and 56% of the last. The failures cluster where the model's mechanism differed from the account we had given of it: the river (E3 and two long-run predictions), disturbance (all four E4 predictions) and the energy budget (E5).

The failures were the productive part. The four E4 predictions failed together and in the same direction, and that pattern, not any single miss, is what sent us to measure the disturbances and find the 0.1%. The two predictions that assumed half lies between the other arms failed, and chasing the disagreement produced the compounding explanation above. The energy prediction failed in the opposite direction from its reasoning: energy did pile up in detritus as expected, but the fixed standing crop makes population more, not less, sensitive. That finding generated the four mid-programme predictions for e45k, three of which then held on an arm that had not finished. The fourth, a population interval of 1,600 to 2,000, missed at 2,066, although the line that produced the interval gave 2,002.

Being wrong on paper costs nothing. Being wrong without having written the prediction down costs the result, because there is then no way to tell a finding from a story told afterwards.

Discussion

Where you stop decides what you find. The learning-rate ratio is 3.7, 1.86 or 1.20 depending on whether the run stops at year 300, 2,000 or 10,000. Each is a correct measurement and each supports a different conclusion. Results about the interaction of learning and evolution in simulated populations should report the trajectory, not a single horizon; even 10,000 years is one horizon, and the inheriting world's standing crop was still rising at the end.

Inheritance as a slow integrator. The clearest effect of inheriting learned values was not on the genes for learning but on what a population can learn at all. A tabular learner averages reward over one life; carried down the generations, the same table averages over thousands of lives and resolves a cost that any single life cannot. Inherited learning has been argued for on speed and solution quality (Ackley and Littman, 1994); here its lasting value is closer to a low-pass filter, raising the signal-to-noise ratio of what a population can learn at all.

Isolation needs a convention to break. The null result in E3 is a statement about the channel, not about isolation. The experiment that would test the explanation gives the channel something arbitrary to settle on: tie one symbol to a nutrient threshold so that it carries information, or make a symbol pay only when a neighbour answers in kind, then compare wall against base.

Limits. Everything here concerns one model with one parameter set; none of it generalises to learning and

evolution in other systems without being tested there. With sixteen seeds and about twenty-two columns per comparison, some t values above 2 are chance, which is why we claim only those above 4. Two of the measures are cruder than they look: the archive-based novelty measure reaches zero by year 300 in every arm, and the count of visited states can outlive the knowledge behind it, as half shows. The ford1 and wall arms ran from copies of the code differing by one line. The predictions were time-stamped in our own files rather than with an external registry; future experiments in this world will be registered externally before they run.

Data and code availability

The full 35-page paper, with every table, the parameter list and the kit's tests, is at Bremner (2026a). Run logs, all 152 checkpoints, the unedited predictions file and the test outputs are at Bremner (2026b). The simulation (664 lines of Python, MIT licence) is at https://github.com/Nerds489/persistent-ai-worlds.

References

Ackley, D. and Littman, M. (1991). Interactions between learning and evolution. In Langton, C. G., Taylor, C., Farmer, J. D., and Rasmussen, S., editors, Artificial Life II, pages 487–509. Addison-Wesley, Redwood City, CA.

Ackley, D. H. and Littman, M. L. (1994). A case for Lamarckian evolution. In Langton, C. G., editor, Artificial Life III, pages 3–10. Addison-Wesley, Reading, MA.

Baldwin, J. M. (1896). A new factor in evolution. The American Naturalist, 30(354):441–451.

Bedau, M. A., McCaskill, J. S., Packard, N. H., Rasmussen, S., Adami, C., Green, D. G., Ikegami, T., Kaneko, K., and Ray, T. S. (2000). Open problems in artificial life. Artificial Life, 6(4):363–376.

Benito, I., Lutzeyer, J. F., and Doerr, B. (2026). A fresh look at Lamarckian evolution and the Baldwin effect. arXiv preprint arXiv:2605.28703.

Bremner, L. (2026a). River world on the desktop. Zenodo.

Bremner, L. (2026b). River world on the desktop: run logs, checkpoints and tests. Zenodo.

Connell, J. H. (1978). Diversity in tropical rain forests and coral reefs. Science, 199(4335):1302–1310.

Cook, J., Lu, C., Hughes, E., Leibo, J. Z., and Foerster, J. (2024). Artificial generational intelligence: Cultural accumulation in reinforcement learning. arXiv preprint arXiv:2406.00392.

Dolson, E. L., Vostinar, A. E., Wiser, M. J., and Ofria, C. (2019). The MODES toolbox: Measurements of open-ended dynamics in evolving systems. Artificial Life, 25(1):50–73.

Hamon, G., Nisioti, E., and Moulin-Frier, C. (2023). Eco-evolutionary dynamics of non-episodic neuroevolution in large multi-agent environments. arXiv preprint arXiv:2302.09334.

Hinton, G. E. and Nowlan, S. J. (1987). How learning can guide evolution. Complex Systems, 1(3):495–502.

Hughes, E., Dennis, M., Parker-Holder, J., Behbahani, F., Mavalankar, A., Shi, Y., Schaul, T., and Rocktäschel, T. (2024). Position: Open-endedness is essential for artificial superhuman intelligence. In Proceedings of the 41st International Conference on Machine Learning, volume 235 of PMLR, pages 20597–20616.

Kirby, S. (2002). Natural language from artificial life. Artificial Life, 8(2):185–215.

Le, N. H. (2026). Organisms write their own evolution: Direct epigenomic inheritance accelerates learning on compositional tasks. arXiv preprint arXiv:2602.00978.

Moreno, M. A., Rodriguez Papa, S., and Ofria, C. (2024). Case study of novelty, complexity, and adaptation in a multicellular system. arXiv preprint arXiv:2405.07241.

Nosek, B. A., Ebersole, C. R., DeHaven, A. C., and Mellor, D. T. (2018). The preregistration revolution. Proceedings of the National Academy of Sciences, 115(11):2600–2606.

Packard, N., Bedau, M. A., Channon, A., Ikegami, T., Rasmussen, S., Stanley, K. O., and Taylor, T. (2019). An overview of open-ended evolution: Editorial introduction to the open-ended evolution II special issue. Artificial Life, 25(2):93–103.

Ray, T. S. (1991). An approach to the synthesis of life. In Langton, C. G., Taylor, C., Farmer, J. D., and Rasmussen, S., editors, Artificial Life II, pages 371–408. Addison-Wesley, Redwood City, CA.

Steels, L. (1995). A self-organizing spatial vocabulary. Artificial Life, 2(3):319–332.

Suarez, J., Du, Y., Isola, P., and Mordatch, I. (2019). Neural MMO: A massively multiagent game environment for training and evaluating intelligent agents. arXiv preprint arXiv:1903.00784.

Watkins, C. J. C. H. and Dayan, P. (1992). Q-learning. Machine Learning, 8(3–4):279–292.

Yaeger, L. (1994). Computational genetics, physiology, metabolism, neural systems, learning, vision, and behavior or PolyWorld: Life in a new context. In Langton, C. G., editor, Artificial Life III, pages 263–298. Addison-Wesley, Reading, MA.